



Understanding Assisted Coding

CSIT205 Generative AI – 2025



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Wollongong
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*SCHOOL OF COMPUTING AND INFORMATION
TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION
SCIENCES*



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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi
Artist Lauren Henry.

The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, partially cut off. The 'U' is at the top, the 'O' is in the middle, and the 'W' is at the bottom.

Everyone is welcome here.

UOW takes pride in
the diversity of our
community.

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uow.info/ally

Subject introduction



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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

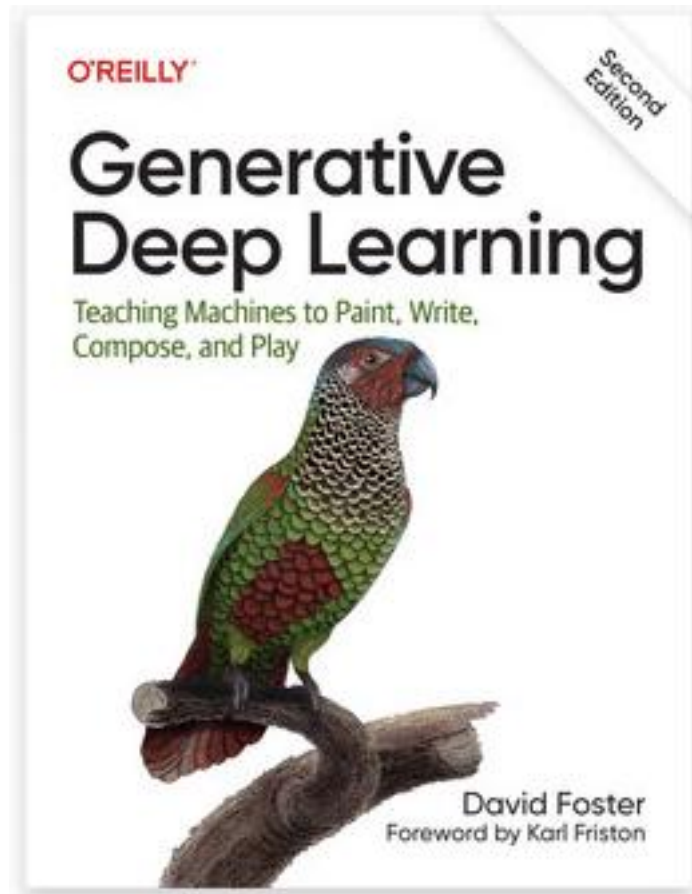
Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Don't forget 

This is the book we're using



<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

Assessment 1

- Deliverables (For both reports, write your full name and UOW ID at the top of first page)
- File name must be in the form of: TXX_NAME_UOWID.txt for prompt-answer pairs where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (not SIM student ID number).

For example, T02_BobChew_8080426.txt

Assessment 1

- Deliverables (For both reports, write your full name and UOW ID at the top of first page)
- File name must be in the form of: TXX_NAME_UOWID.doc for report where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (not SIM student ID number).
For example, T02_BobChew_8080426.doc

Assessment 2

Assessment 2 – Generative AI business case

- (Day 1 – Form a group of 5 or 6 students and select a responsible student as a group leader. The group leader of a tutorial class is to inform the lecturer of the tutorial class, group members UOW id and names before end of 8th day.)
- (All team members must be present unless there is a valid reason during the interview with CTO during lab 3 and presentation during lab 4.)

Assessment 2

Assessment 2 – Generative AI business case

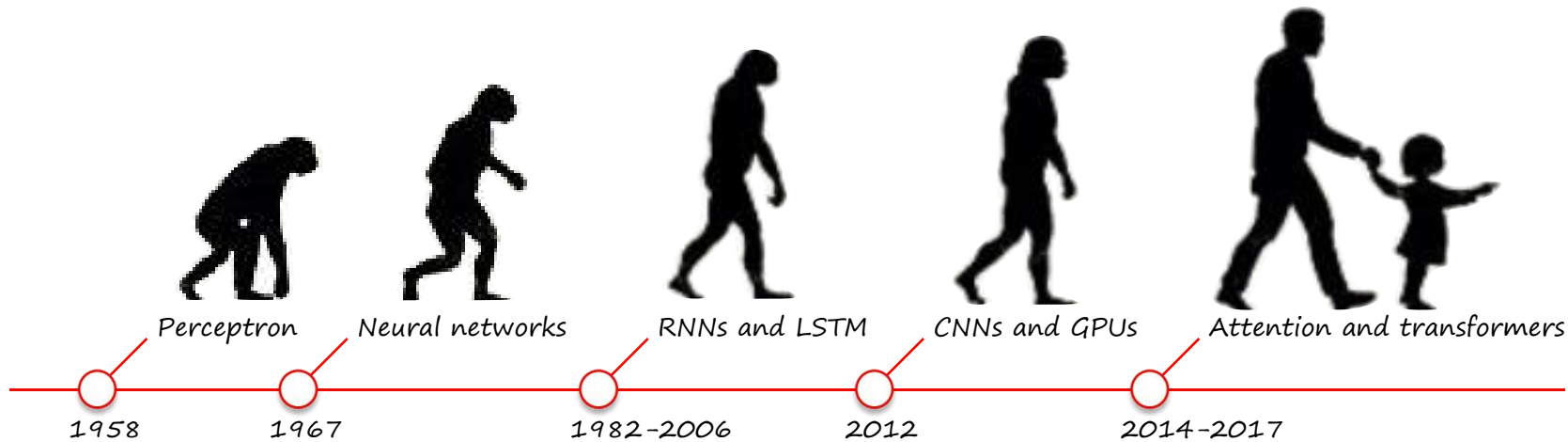
- File name must be in the form of: TXX_NAME_UOWID.doc where XX is your tutorial group, NAME is the group leader full name (without space or underscore) and UOWID is your 7-digit UOW ID number (not SIM student ID number).
For example, TO2_BobChew_8080426.doc
- (For the report and presentation slides, write the full name of each group member and their UOW ID at the top of first page)

What did we discuss last week?



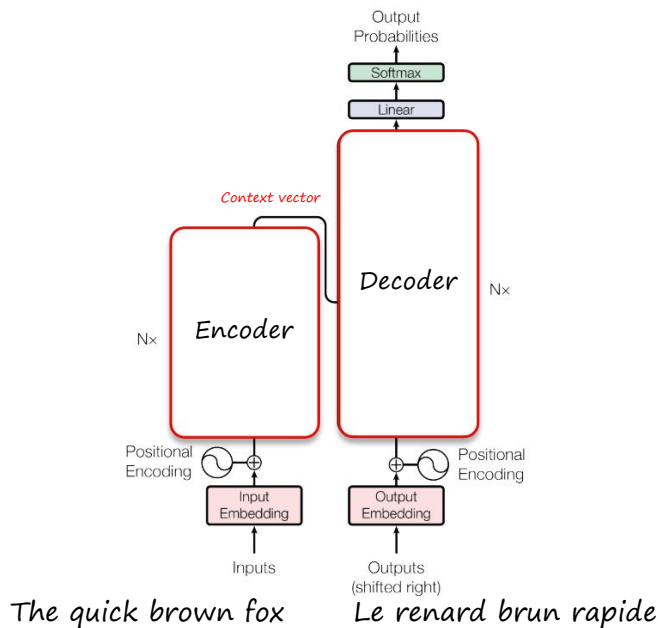
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Attention and transformers



Transformer model

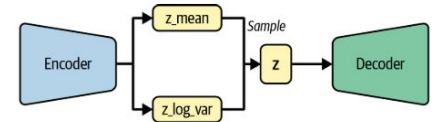
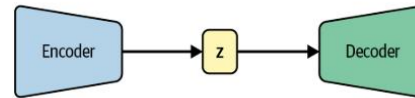
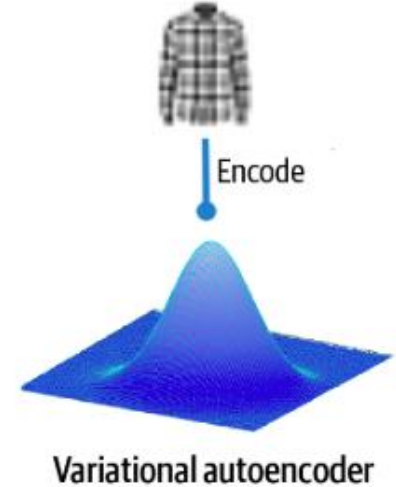
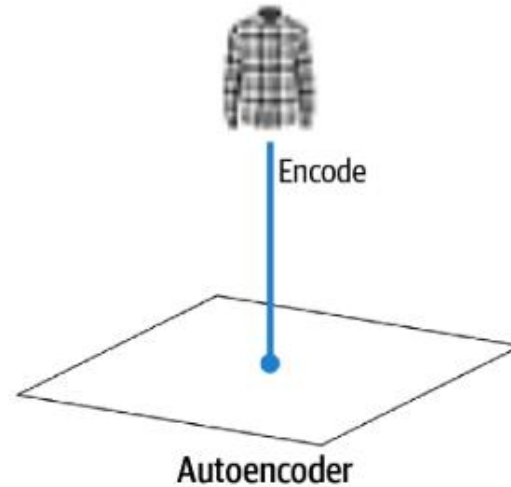
- 'Attention is all you need' (Vaswani et al., 2017)
- Uses self-attention instead of traditional recurrent or convolutional architectures.
- Encourages parallelisation by allowing multiple heads to attend simultaneously.
- Typically consists of an encoder and decoder, with a multi-layer



The Encoder maps the input to a limited dimensional space ('context vector').
The Decoder generates output (given the context).

Variational Autoencoders

- Instead of modelling the learned image as a single spot, model it as a normal / Gaussian distribution.
- So instead of a single vector, we need a mean vector and variance vector.
- We then sample from the two vectors.



Why are we here?

CSIT205 Generative AI - Module 1

AI assisted coding

- Image generation part 2
- AI-assisted coding

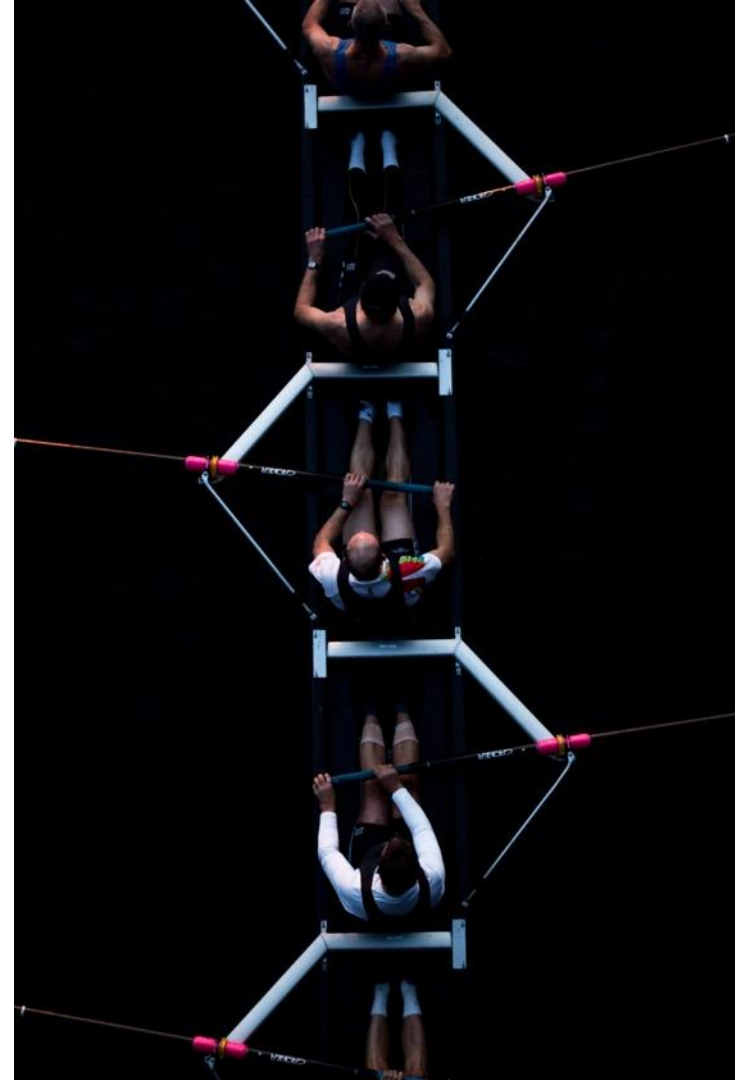
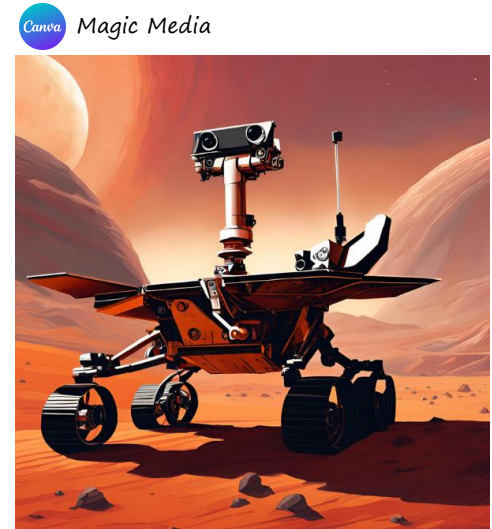
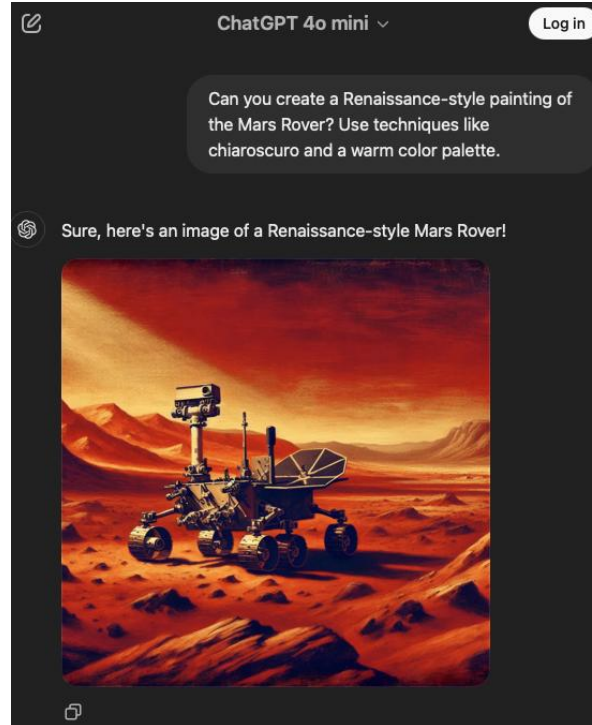
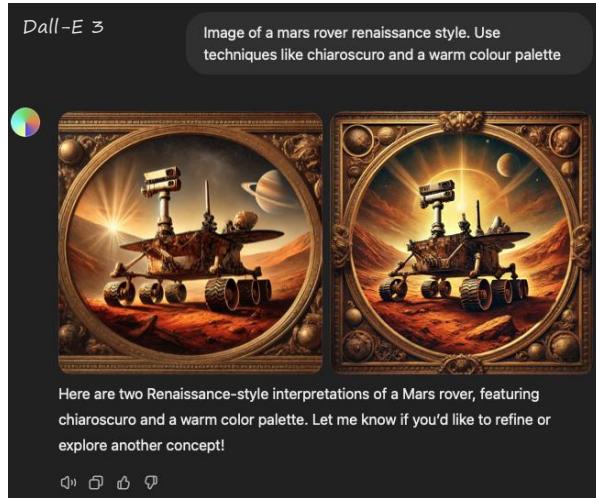


Image Generation part 2



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How we know Image Generation



Diffusion Models



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Advantages and Applications of Diffusion models

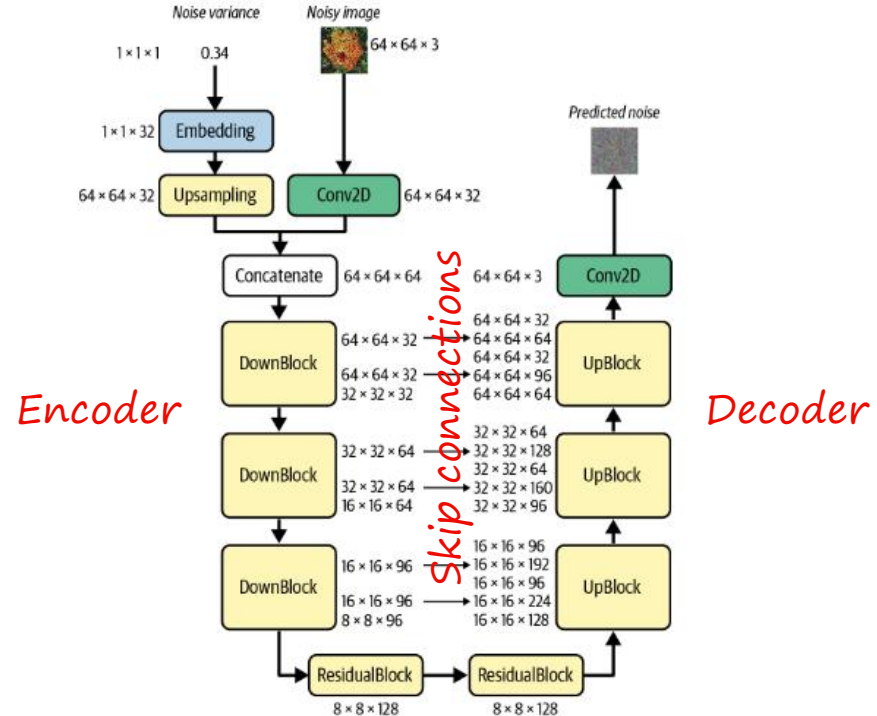
- *High-fidelity generation: can produce realistic and detailed images. Originally used for image segmentation.*
- *Mode coverage: can capture a wide range of modes in the data distribution.*
- *Robustness: can be trained to be robust against adversarial attacks and noise.*



'Unicorn programming behind a laptop' generated by Dall-E 3

Diffusion models

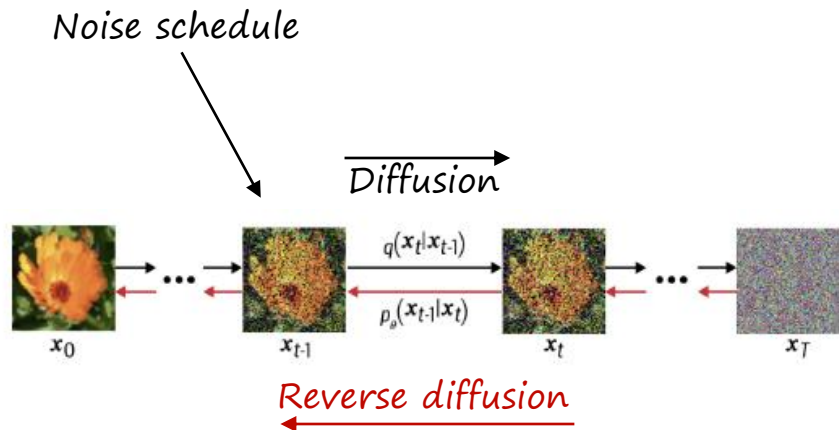
- Unsupervised learning framework
- Uses diffusion and reverse diffusion to learn how to reconstruct an image
- Can learn complex distributions, including those with noise and uncertainty
- Key mechanisms are the encoder, decoder, and skip-connections



U-Net (Foster, 2003, p.217)

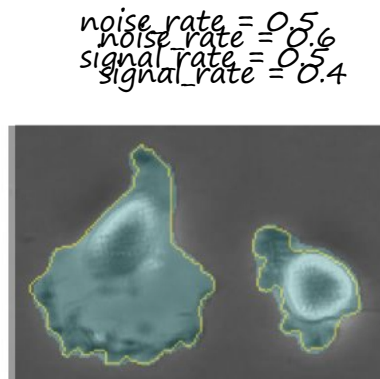
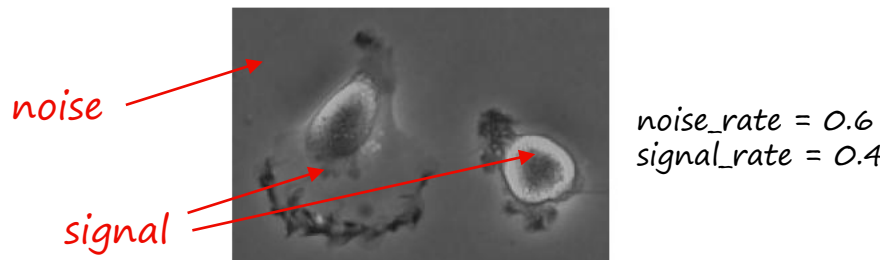
Key components

- *Noise schedule: controls the rate at which noise is added to the data sample*
- *Diffusion process: iteratively refines the data sample through multiple steps*
- *Reverse diffusion: reverses the diffusion process to generate new data samples*



How to train Diffusion models

1. Add noise to training images:
noise_rate and signal_rate.
2. Train the model to predict noise_rate
and signal_rate.
3. Denoise the noisy image.
4. Calculate the loss.
5. Update model weights.



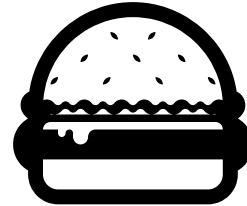
noise_rate_loss = 0.1
signal_rate_loss = 0.1

10-minute Break!

This will help you learn



10 minutes



How do you control the rate at which noise is added to a diffusion model?



Noise schedule



What is diffusion?

- A. The process of generating new data samples.
- B. The process of adding noise to images during the training process.
- C. The concept of training a model to generate new data samples.
- D. The process of updating weights to minimise the loss

AI-assisted Coding



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Challenges of manual coding

- ✗ *Time-consuming and labour-intensive*
- ✗ *Error-prone and prone to bugs*
- ✗ *Limited scalability and maintainability*



image of a frustrated software developer in the style of the scream by Edvard Munch, MS Copi

Types of AI assisted coding

- *Code analysis and review: AI identifies bugs and suggests fixes, enhancing the debugging efficiency.*
- *Code completion and suggestion: AI tools suggest code snippets and auto-complete lines of code to speed up the coding process.*
- *Code refactoring and optimisation: AI can suggest and automate refactoring processes to improve code structure without changing its functionality.*

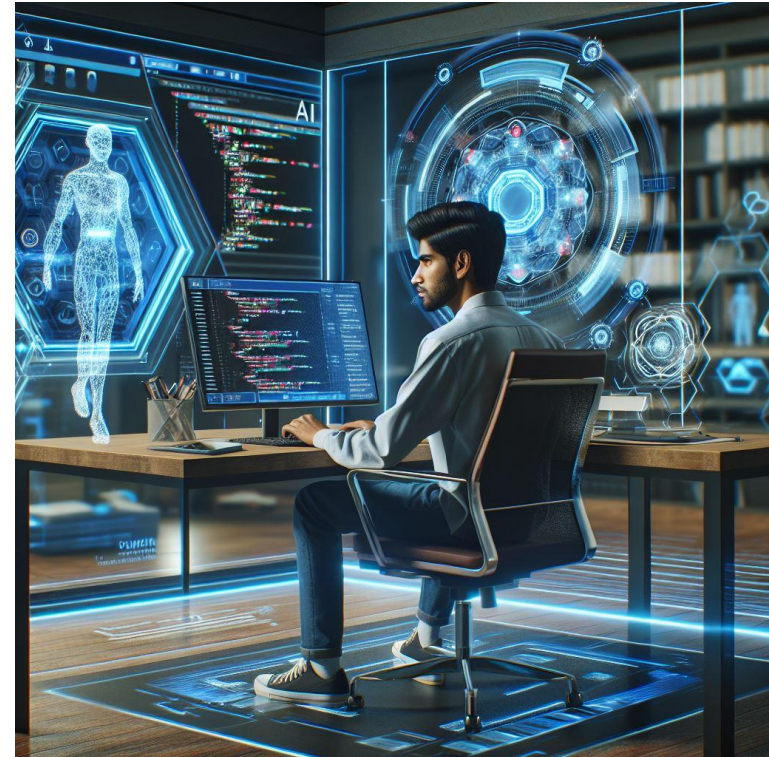
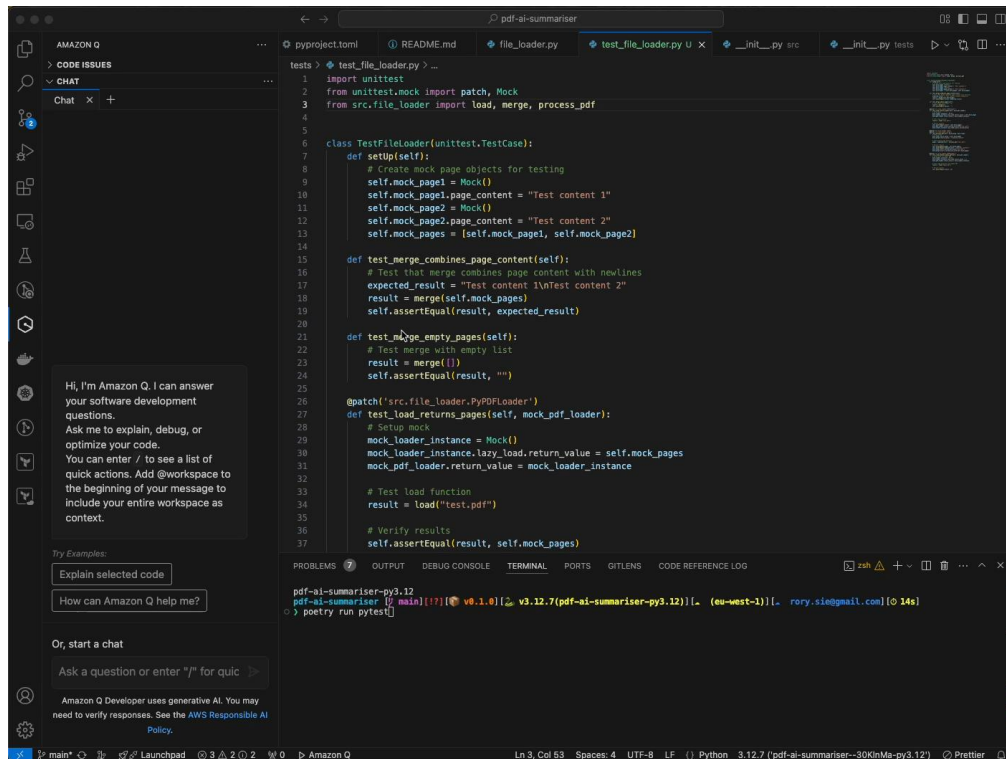


image of a futuristic coding environment with AI-inspired tools and interfaces, MS Copilot

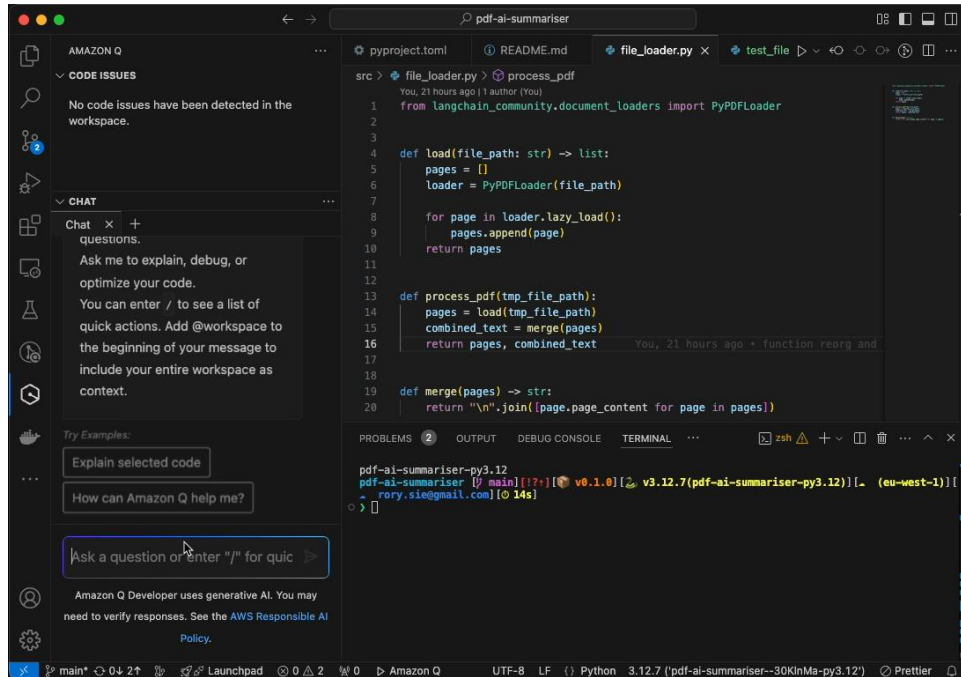
Code Analysis

- Analyse the quality of code, e.g. functions, directory structure.
- Provide suggestions on how to improve.



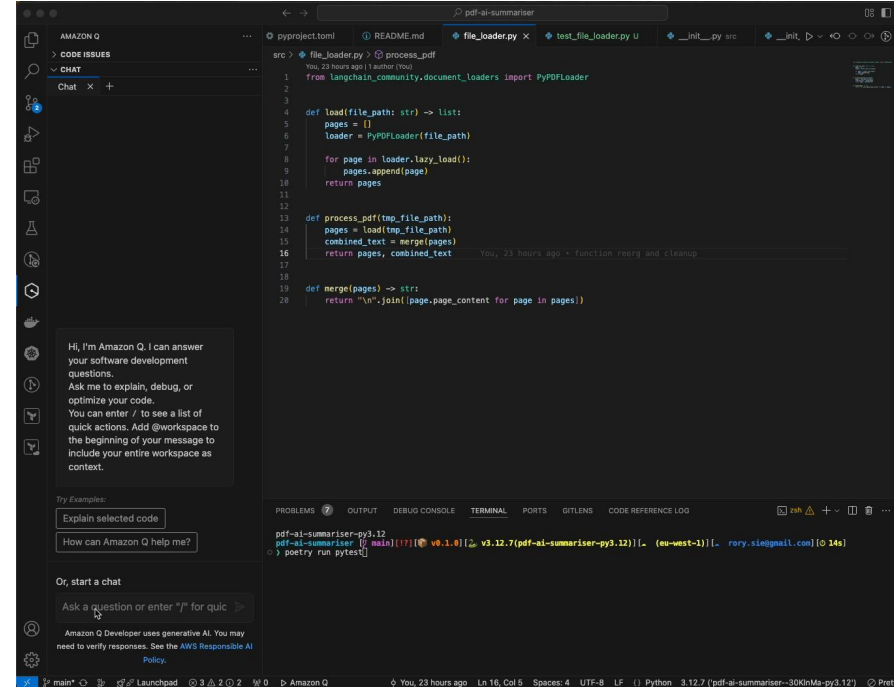
Code Completion

- Suggest code completion
- Translate natural language to code
- Write unit tests



Code Refactoring

- *Convert repetitive code into reusable code*
- *Modernise code, e.g. implement new language features.*
- *Performance optimisation*
- *Reduce code complexity, e.g. break down complex chunks of code.*
- *Security enhancements*
- *Error handling*

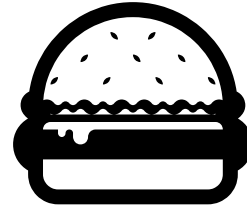


10-minute Break!

This will help you learn



10 minutes



What types of AI-assisted coding do we have?



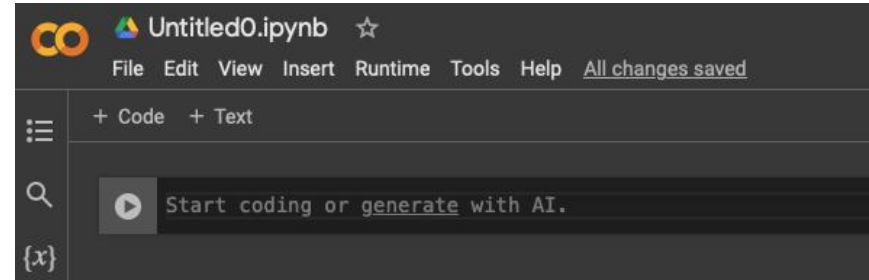
- *Code completion*
- *Code analysis*
- *Code refactoring*



10 minutes

It's your turn to use AI-assisted coding!

1. Go to colab.google.com.
2. Choose 'New notebook'.
3. Click generate with AI to start writing.
4. Ask AI to create a data frame with 10 student marks and grades.
5. Press the 'play' button to execute the code.
6. What happened? How can we optimise this?
7. Scroll down and click 'View recommended plots'. Choose one that shows the grades.
8. Click 'Show code' to see the code that generated the plot.



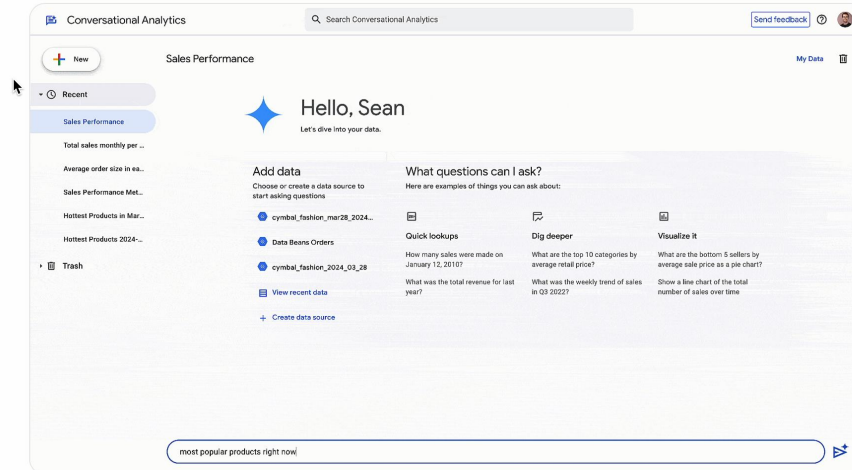
U GenAI and Data Analytics O W



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Data Visualisation and Exploration

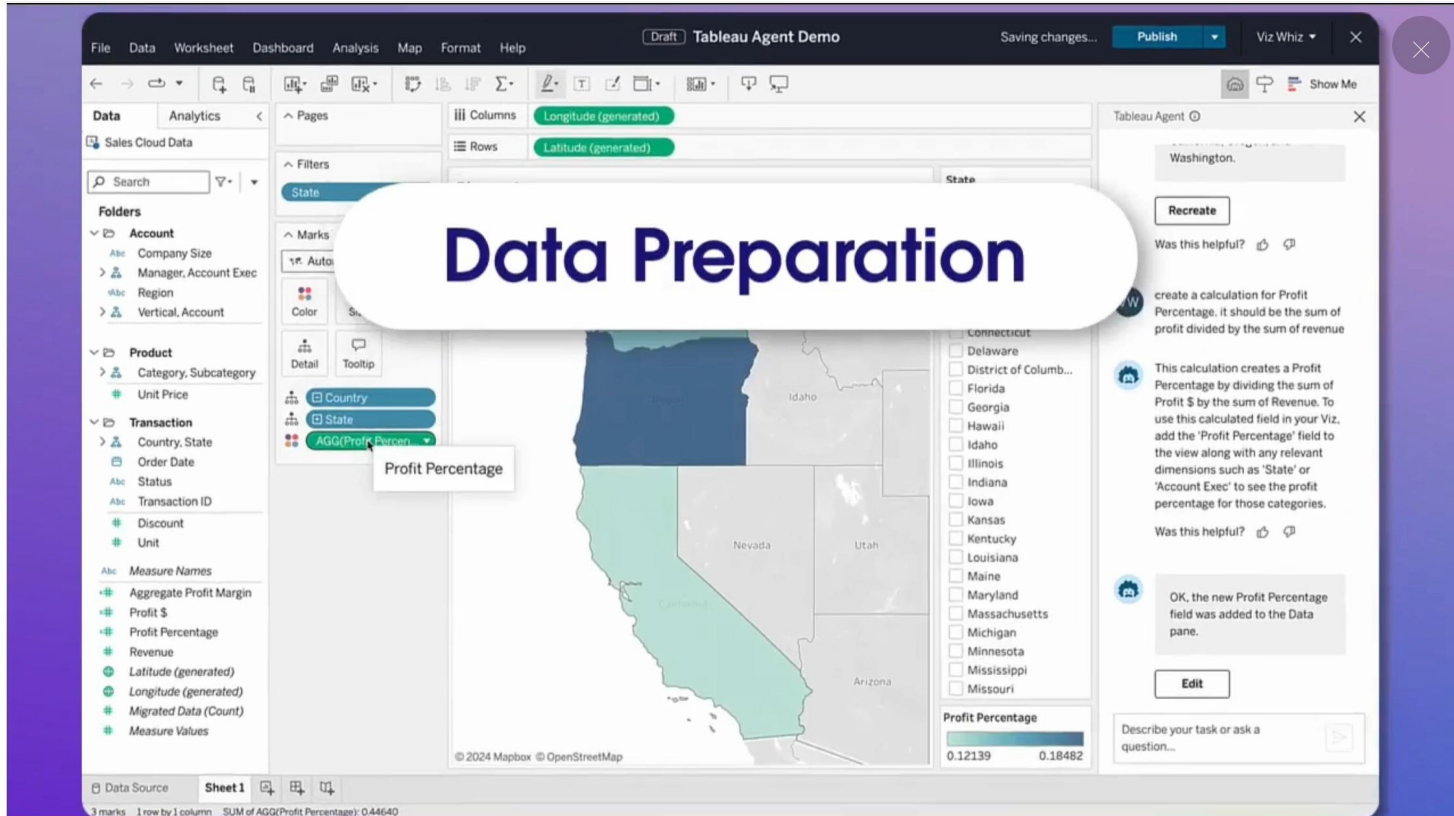
semantic models available from Looker, we establish a single source of truth for your data, providing consistent metrics across your organization. Now, your entire company, including business and analyst teams, can chat with your data and obtain insights in seconds, fully enabling your data-driven decision-making culture.



You can leverage Conversational Analytics, using Gemini in Looker, to find top products, sales details, and dive deeper into the answers with follow-up questions.



Data Processing



Data Processing: SQL generation

 QueryGPT Beta Workspaces ? X

 Pro Tip

If your use case is specific, consider creating a custom workspace to improve accuracy.

 Hey Jeffrey, ask me a question, and I can help you by generating a first-draft query.

Find the number of trips completed in Seattle yesterday



 I'm planning to use the tables below to generate the query

Tables to be used:

uber.trips

uber.cities

 Looks Good (0:08)

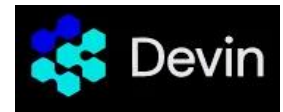
```
{
  "explanation": "To answer the question, we need to join the 'uber.trips' table with the 'uber.cities' table to filter trips by the city name 'Seattle'. We will use the 'city_id' column to join these tables. Since 'uber.trips' is partitioned by 'date_val', we must include a filter for this partition column. We will filter for trips that were completed yesterday using the 'is_trip_completed' column and the 'date_val' column. The 'date_val' column will be cast to a DATE type to compare with the date function that calculates yesterday's date. We will use a COUNT() function to get the number of trips and GROUP BY to group the results by 'city_name'.",
  "sql_query": "SELECT COUNT(*) AS number_of_trips FROM uber.trips ft JOIN uber.city dc ON ft.city_id = dc.city_id WHERE ft.is_trip_completed = TRUE AND dc.name_of_city = 'Seattle' AND DATE(ft.date_val) = DATE_ADD('day', -1, CURRENT_DATE) AND ft.date_val IS NOT NULL"
}
```

Benefits of AI-assisted coding

- ✓ *Increased productivity and efficiency*
- ✓ *Improved code quality and maintainability*
- ✓ *Reduced errors and bugs*
- ✓ *Enhanced collaboration and teamwork*

AI assisted coding tools

- *GitHub Copilot (paid)*
- *Tabnine (free tier)*
- *Amazon Q Developer (free tier)*
- *Codeium (free tier)*
- *ChatGPT (free tier)*
- *Cursor (free tier)*
- *Gemini Code assist (paid)*
- *Special mention: Devin (US\$500)*



Lab A

- *In Lab A you will use AI-assisted coding more extensively under the guidance of a tutor.*
- *Lab instructions have been made available*
- *If you choose not to attend, unenrol from the Lab*

Wrap-up



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Summary

- *Diffusion models are trained by adding noise to images and reverse diffusion to reconstruct an image.*
- *We broadly have three types of AI-assisted coding: code completion, code analysis, and code refactoring.*

For next lecture

- *Read Foster, 2023, Chapter 10/13*

Take Home Message...

*Consider the message that you want to create when
designing your interfaces.*



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Things to think about

What is the purpose of prototyping?

*What is the difference between Lo-fi
and Hi-fi designs?*

*What is a storyboard? When can
these be used?*

What should you show your client?

References and Resources

- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation (No. arXiv:1505.04597). arXiv.
<https://doi.org/10.48550/arXiv.1505.04597>

Questions



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